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Paleomagnetic data from the Alaska Peninsula

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ABSTRACT

Recently collected paleomagnetic data, combined with earlier published data, tend to confirm earlier models for the tectonic development of southwestern Alaska. The models involve a basically northward migration and rotation of southwestern Alaska between Cretaceous time and the present, with an earlier southward motion. If the paleolongitudes of the reconstructions are constrained by always keeping southwestern Alaska in contact with North America, some geologic conflicts arise. These may possibly be resolved if large-scale en echelon movement can be proved along such features as the Bruin Bay and Border Ranges faults.

INTRODUCTION

Until a decade or so ago, the geology of Alaska was well known only for small patches, with the remainder known only on a reconnaissance basis. In the last few years, in part because of the need to know more about the fossil fuel reserves of the area and in part because of the availability of helicopter logistic support, a picture of the overall geologic situation has emerged. However, the only hard and fast conclusion that can be drawn from this increase in knowledge so far is that Alaska's tectonic history has been very complex and that simplistic models such as the oroclinal bending hypothesis of Carey (1958) have very limited usefulness.

In terms of the paleomagnetic data discussed below, only the post-Paleozoic tectonic history is considered, and, as the only paleomagnetic data available are from south-central and southwestern Alaska, only these areas can be considered directly. However, the data have implications for other parts of Alaska and the northeast Pacific.

GEOLOGIC SETTING

Along the Alaska Peninsula a series of Mesozoic and Tertiary sedimentary sequences and volcanic and intrusive complexes are exposed. These rocks appear to have been formed in several island-arc and arc-trench gap situations and provide the opportunity to study paleomagnetically the movements of what appears to be a geologically coherent block. Burk (1965) presented an excellent summary of the geology of the Alaska Peninsula; his work is the source of most of the descriptions of geologic settings in the following paragraphs.

The sedimentary sequences, which are of most interest for the paleomagnetic studies presented in this paper, and the early Mesozoic batholithic complexes are for the most part separated by the Bruin Bay fault (Fig. 1). This fault runs nearly the entire length of the Alaska Peninsula and is a high-angle dip-slip fault that has allowed uplift of the batholiths on the northwest side. These batholiths became the predominant source of material for the sedimentary sequences on the southeast side of the fault. Evidence for subsequent middle and late Tertiary volcanism and intrusive activity and Quaternary volcanic activity is seen on both sides of the Bruin Bay fault.

The oldest rocks exposed on the Alaska Peninsula are of Permian-Triassic age and are composed of predominantly detrital volcanic rock, extrusive flows, limestone, and chert. The younger Jurassic and Cretaceous sequences of sandstones, siltstones, and conglomerates are exposed in scattered locations along the length of the peninsula. The Lower Jurassic volcanic and volcanic-rich sediments are not present in the southwest part of the peninsula, but a very complete section of Middle and Upper Jurassic sedimentary rock is well exposed in the Tuxedni Bay region to the northeast, where it conformably overlies Lower Jurassic rocks. Part of this section, the Middle Jurassic Tuxedni group, is composed of siltstone, arkose, graywacke sandstone, conglomerate, massive sandstone, and shale. A total of at least 3,000 m (10,000 ft) of the combined Tuxedni group and the overlying upper Middle Jurassic Chinitna Formation are exposed at Tuxedni Bay (Detterman and Hartsock, 1966).

The Upper Jurassic Naknek Formation conformably overlies the Chinitna Formation and is composed of a basal conglomerate, arkosic sandstone, and siltstone. Exposed the entire length of the Alaska Peninsula, the Naknek Formation forms a distinctive marine rock type that reaches its maximum thickness of 3,000 m (10,000 ft) in the Wide Bay area. Southwest of Wide Bay the Late Jurassic–Early Cretaceous Staniukovich Formation conformably overlies the Naknek. The Staniukovich Formation, composed of fine-grained feldspathic sandstone and arkose, reaches a maximum thickness of 600 m (2,000 ft) in the Wide Bay area and varies irregularly to the southwest. In the Cape Douglas area the Staniukovich Formation is conformably overlain by the Lower Cretaceous Herendeen Limestone.

The Middle Jurassic through the Lower Cretaceous section represents sedimentation in a rapidly subsiding area close to an eroding granitic source. Uplift and minor deformation occurred on the Alaska Peninsula in middle Cretaceous time, creating an unconformity between Upper and Lower Cretaceous rocks. The Upper Cretaceous section, exposed predominantly southwest of Wide Bay, is composed of a thick basal coal member and argillaceous sandstone, siltstone, and shale.

The Tertiary sequence on the Alaska Peninsula is characterized by an almost complete rock record, in places thousands of metres thick, consisting largely of basal-

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tic and andesitic flows and sills and their sedimentary debris. Paleocene rocks overlie with slight unconformity the Upper Cretaceous or older rocks, indicating minor uplift and erosion. During earliest Tertiary time a quartz diorite was intruded into Upper Cretaceous slate and graywacke along a line from the Shumagin Islands through Kodiak Island and on to the Kenai Peninsula (Kienle and Turner, 1976). This perhaps corresponds to the episode of slight uplift in early Paleocene time. The slight deformation and uplift that apparently occurred in late Oligocene–early Miocene time may correspond to the intrusion of middle-Tertiary granodiorite bodies along the Pacific coast of the Alaska Peninsula. Anticlinal and synclinal folding and the local faulting associated with these anticlinal complexes occurred during Pliocene time. From Kamishak Bay northeast along the west coast of Cook Inlet, a narrow strip of land southeast of the Bruin Bay fault has undergone tight folding and minor faulting.

From Kamishak Bay southwest to Becharof Lake, folding is reduced to a broad anticline, the Kamishak anticline, and minor gentle folds that have a general homoclinal dip toward Shelikof Strait. Northwest of the Bruin Bay fault, folding is local and generally restricted to rocks of early Mesozoic age and is probably due to the intrusion of the Jurassic Aleutian Range batholiths.

Southwest of Becharof Lake the style of folding changes. Three long anticlinal arches can be seen: one from Puale Bay to the Aniakchak Crater area, one from Chignik Bay to Stepovak Bay, and one from Port Moller to Cold Bay. All are broad features with two smaller anticlines and minor high-angle reverse faults on the southern margin.

Volcanism has continued from Pliocene time to the present and is continuing along the length of the Alaska Peninsula. Quaternary and recent flows and associated pyroclastic material, glacial material, and stream deposits make up the recent rock record on the Alaska Peninsula.

PALEOMAGNETIC TECHNIQUE

All the new data presented in this paper were obtained using the same basic procedures. An area for sampling was selected on the basis of the known geology and also the accessibility. Core samples were then drilled from as many individual units in the area as time and exposure allowed. Each unit, be it

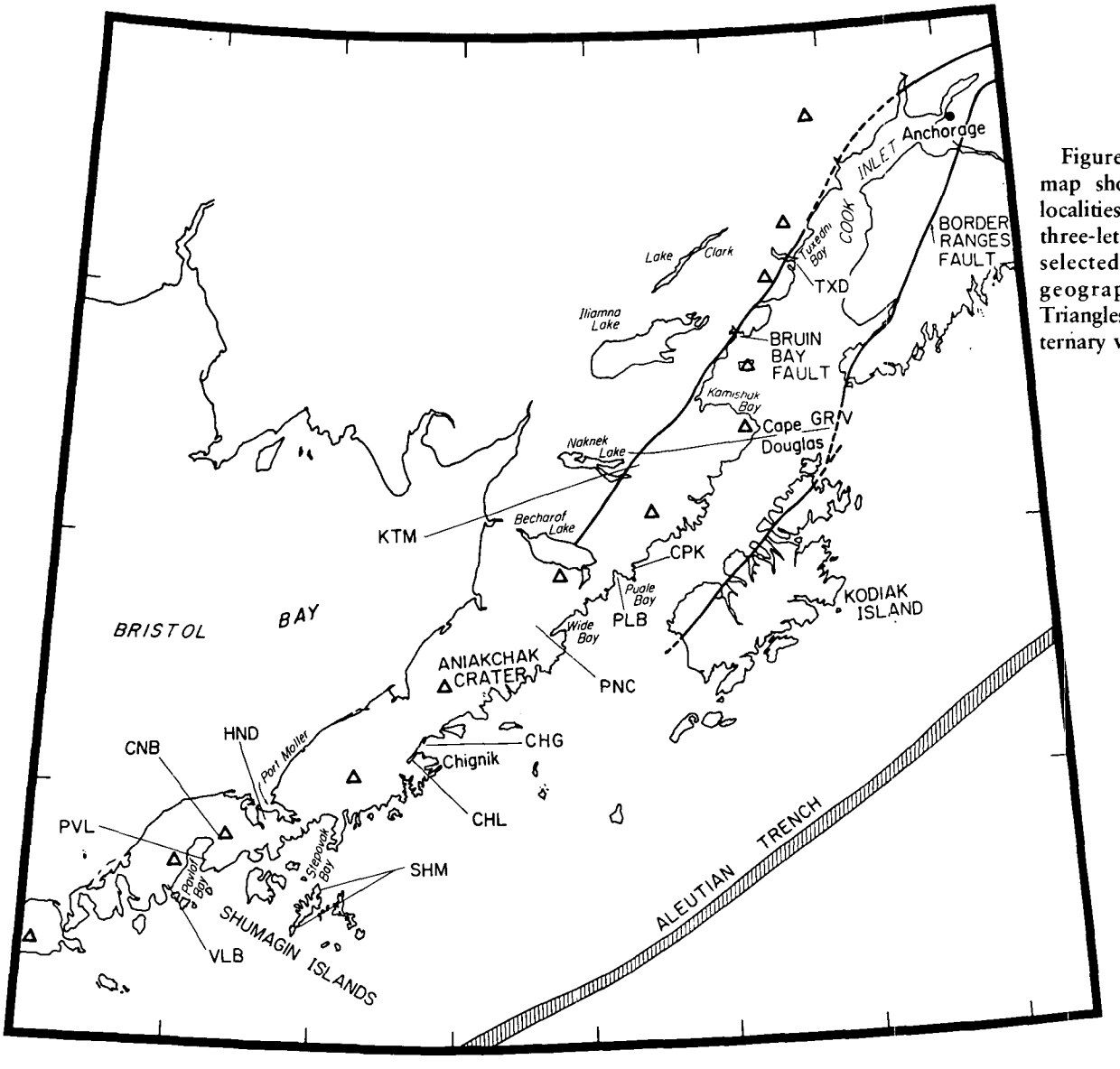


Figure 1. Location map showing sampling localities, represented by three-letter codes, and selected geologic and geographic features. Triangles represent Quaternary volcanoes.

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a sedimentary bed, lava flow, or dike, is considered a site, using the terminology of McElhinny (1973), who considered a site to be a discrete unit of time. In the few cases where more than one core was taken from the same horizon, the mean of these is considered as a site.

All the cores were drilled from in situ outcrops and oriented using a sun compass, by sighting on geographic features, and by magnetic bearings, or as many of these as conditions allowed. At no sampling area was the orientation exclusively magnetic, so for all cases where a magnetic bearing was the only system available for a particular core, a local reading of the magnetic declination was available from the orientation data for nearby cores. Similarly, comparisons of the magnetic declinations for several nearby cores give an estimate of any errors introduced by very local declination changes. These were commonly found to be negligibly small for each individual sampling area, although the declination showed significant changes from one area to another.

Each of the 2.5-cm diameter cores was sliced into discs 2 cm thick. The lowermost of these discs was used for the magnetic measurements in an attempt to minimize the effects of surface weathering.

The direction and intensity of the remanent magnetization of the samples were measured using both a Schonstedt SM-1 spinner magnetometer and an ScT superconducting magnetometer. Following the measurement of the natural remanent magnetization (NRM), the samples were remeasured after demagnetization in progressively higher alternating magnetic fields (AF). The demagnetizing field was increased to a point where the intensity of magnetization was severely reduced or the scatter of the magnetic vector directions became essentially random.

SELECTION OF DATA

In only two cases could the commonly used technique of selecting the "best" demagnetization level as that for which the scatter of individual site directions was a minimum be applied, because in general the individual sites exhibited different magnetic behaviors and so had no common best AF level. This presumably results from the wide range of rock types sampled in most areas.

A combination of several methods was eventually used to select a best estimate of the original paleomagnetic field direction. Some care was taken in this selection procedure, because the basic unselected data appear widely scattered. In particular, great care was taken not to introduce a personal bias toward a preferred direction; therefore, all the data were selected before applying any local bedding or regional dip corrections. In this way, the location of any group of magnetic directions, as plotted on a stereographic projection, would have no obvious meaning with regard to a paleomagnetic field direction. In cases where the bedding is horizontal, a spurious dip correction could be applied to all the data before selection and removed later.

It was found that when the changes in direction with demagnetization for each site were plotted together, there was generally a tendency for the magnetic vectors to migrate toward a particular area of the stereographic plot before dispersing as their magnetic intensities decreased. The "focus" of this migration was often also associated with a tighter clustering of a few individual data points. The center of this clustering was then estimated by eye. The curves showing intensity of magnetization against demagnetizing field were also plotted. In most cases, these showed one or more inflection points assumed to be due to the removal of a particular magnetic component. On this assumption it was possible to pick out the range of demagnetization levels with the highest probability of having removed secondary magnetic components. It was often found that the magnetic vector direction associated with these selected AF demagnetization levels also corresponded quite closely to the closest approach to the estimated mean of the clustered points mentioned above. In these cases, this data point was selected as representing the ancient field. Where the demagnetization curves for individual sites did not indicate an obvious choice of demagnetization level, the level at which the vector approached closest to the estimated mean was used.

The next step was to take the mean of all the selected data points for the sampling area in question, reject any that lay outside a circle with an arbitrary radius of 50°, and recompute the mean and the statistics of the distribution about the mean (Fisher, 1953). These selected data means, together with their statistics and the number of sites rejected in each case, are listed in Table 1. An example of the selection process is also shown in Figure 2.

In addition to using the method described above, we also noted the direction of the measured magnetic field directions relative to the present field. In most cases it was found that they were significantly different from one another.

After the selection process, the data were converted from magnetic vector directions relative to present geographic coordinates to vector directions relative to present north and ancient horizontal. This latter conversion assumes that the observed dip of the bedding was the result of rotation about a single horizontal axis. It is hoped that any errors introduced by this assumption will be averaged out by the use of a large number of sampling areas with a wide geographic spread; however, it is possible that a systematic error could be introduced if two or more periods of very large-scale regional deformation occurred. These errors do not, of course, affect the inclination of the vector (which determines the paleolatitude), but only the declination.

The final vector direction was then converted to the equivalent virtual geomagnetic pole (VGP), which describes the location of the axis of the geocentric dipole needed to produce the observed vector. These VGP values are listed in Table 2. Virtual geomagnetic pole positions are independent of site latitude, thus allowing direct comparison of data from area to area.

SELECTION OF DATA FOR PERIOD MEANS

To obtain the final mean VGP values for the Tertiary, Cretaceous, and Jurassic Periods for the areas sampled, the data were further selected by rejecting all groups of data or area means where the best estimate of the Fisher parameter k was less than 6 (Fisher, 1953), and by rejecting any mean lying excessively far from the mean of the remaining means. This latter selection criterion is deliberately vague, as the distribution of the area means is very different for different ages, but, in general, anything more than 50° from the area mean was rejected. The final selection is admittedly somewhat subjective; the reasons for each selection are explained for the individual cases.

AGE CONTROL

In many parts of Alaska the dating of rocks is difficult, and in many instances dramatic changes in assigned ages have occurred within the last several years. Perhaps the most dramatic, with regard to models of the tectonic development of the Alaska Peninsula and Aleutian system, was the discovery that the "Permian" sedimentary

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TABLE 1. PALEOMAGNETIC DATA

Area	Site no.	VGP λ	VGP φ	D	I	Polarity	AF (Oe)	Intensity (× 10^{-6} emu/cm^3)
HND-1	1000	18	81	303	1	+	100	1.7
lat 55°75′N, long 199°2′W;	1001	35	130	276	59	+	100	3.3
thickness sampled 125 m;	1002	23	80	306	9	+	100	44
bedding correction: azimuth	1003	39	82	313	34	+	100	57
280°, dip 17°; mean of	1004	32	109	289	44	+	100	14
12 sites; VGP dispersion	1005	40	122	287	58	+	100	20
parameter 16.2, cone of	1006	0	102	276	− 9	+	100	94
confidence 10.8°; no sites	1007	36	78	314	28	+	100	91
rejected	1008	10	83	297	−11	+	100	52
	1009	16	97	287	14	+	100	164
	1010	33	75	315	21	+	100	386
	1011	34	84	309	29	+	100	142
	Mean:	27	93	298	23			
HND-2	1013	76	186	350	79	+	400	2
lat 55°8′N, long 199°2′W;	1014	74	37	354	59	+	400	7
thickness sampled 41 m;	1016	57	18	1	40	+	400	8
bedding correction: azimuth	1017	62	106	318	64	+	400	2
347°, dip 18.5°; mean of	1019	57	352	16	43	+	400	3
7 sites; VGP dispersion	1021	44	99	305	49	+	50	18
parameter 7.1, cone of confidence	1022	55	295	49	59	−	400	4
24.3°; 3 sites rejected								
	Mean:	77	40	354	60			
HND-3	1023	43	198	184	89	+	100	15
lat 55°8′N, long 199°3′W;	1024	59	139	302	73	+	100	13
thickness sampled 22 m;	1025	56	128	300	68	+	50	16
bedding correction: azimuth	1026	64	103	321	69	+	50	13
215°, dip 25°; mean of	1027	40	277	73	58	+	200	8
12 sites; VGP dispersion	1028	72	277	32	71	+	50	18
parameter 5.5, cone of	1030	68	145	319	76	+	200	10
confidence 20.4°; 3 sites	1031	38	263	85	63	+	100	16
rejected	1032	66	233	44	80	+	100	9
	1033	39	260	86	65	+	200	6
	1034	34	279	76	52	+	200	3
	1036	63	152	307	77	+	400	4
	Mean:	70	223	33	80			
HND-4	1039	23	337	38	− 6	−	100	4
lat 55°7′N, long 199°3′W;	1040	4	332	51	−34	+	200	1
thickness sampled 34 m;	1041	39	41	343	13	−	100	3
bedding correction: azimuth	1044	17	68	314	−12	+	100	5
112°, dip 28°; mean of	1046	59	328	28	51	−	200	2
8 sites; VGP dispersion	1047	21	57	325	12	−	200	2
parameter 4.1, cone of	1049	19	13	6	−29	+	200	2
confidence 31.4°; 5 sites	1050	36	318	47	28	−	400	2
rejected								
	Mean:	34	8	7	3			
HND-5	1054	40	295	61	48	−	100	4
lat 55°8′N, long 199°1′W;	1055	30	263	92	59	−	200	7
thickness sampled 52 m;	1056	39	284	69	54	+	200	1
bedding correction: azimuth	1057	39	287	67	51	−	400	2
232°, dip 40°; mean of	1058	40	300	58	45	−	50	2
7 sites; VGP dispersion	1059	37	295	63	46	−	200	3
parameter 60.2, cone of	1061	36	292	66	47	+	400	2
confidence 7.8°; 5 sites								
rejected	Mean:	38	288	67	50			

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TABLE 1. *(Continued)*

Area	Site no.	VGP λ	VGP φ	D	I	Polarity	AF (Oe)	Intensity (× 10^{-6} emu/cm^3)
HND-6	1064	37	90	306	37	−	100	2
lat 55°7′N, long 199°3′W;	1066	12	90	292	− 2	−	100	7
thickness sampled 41 m;	1067	6	78	300	−23	−	400	3
bedding correction: azimuth	1068	10	123	264	30	−	200	1
219°, dip 24°; mean of	1069	13	85	297	− 3	−	100	2
8 sites; VGP dispersion	1070	21	87	300	10	−	200	4
parameter 13.5, cone of	1072	14	79	303	− 8	−	100	3
confidence 15.6°; 4 sites	1073	−22	106	259	−36	−	400	3
rejected								
	Mean:	11	92	291	1			
HND-7	1075	40	76	318	31	−	400	2
lat 55°7′N, long 199°3′W	1076	24	78	308	9	−	400	1
thickness sampled 158 m;	1077	68	119	321	70	+	400	1
bedding correction: azimuth	1080	30	105	291	39	−	400	1
201°, dip 24°; mean of	1081	29	45	337	4	−	400	2
7 sites; VGP dispersion	1083	8	125	262	29	−	400	1
parameter 7.4, cone of	1084	31	100	295	37	+	400	0.5
confidence 23.7°; 4 sites rejected								
	Mean:	35	91	304	33			
HND-8	1086	2	112	269	6	−	200	20
lat 55°8′N, long 199°2′W;	1088	29	99	295	34	+	100	3
thickness sampled 88 m;	1089	−10	121	254	− 3	+	100	24
bedding correction: azimuth	1090	43	63	328	29	+	200	29
89°, dip 24°; for sites	1091	30	105	291	38	+	50	28
1086 to 1094, mean of 7	1092	46	110	300	57	+	200	47
sites, VGP dispersion	1094	26	102	291	33	+	200	41
parameter 10.0, cone of confidence 20.1°, 2 sites rejected; for sites 1095 to 1100, mean of 6 sites, VGP dispersion parameter 8.5, cone of confidence 24.5°, no sites rejected	Mean:	25	103	288	30			
	1095	14	260	106	46	+	400	0.4
	1096	7	287	87	14	+	400	0.9
	1097	47	267	73	66	+	400	0.6
	1098	43	230	109	78	+	100	1
	1099	33	234	118	72	+	200	1
	1100	59	226	67	82	+	100	1
	Mean:	36	254	93	61			
HND-9	1101	42	68	324	+31	−	200	5
lat 55°7′N, long 199°2′W;	1102	6	51	320	−40	−	100	3
thickness sampled 23 m;	1103	32	105	293	41	−	400	2
bedding correction: azimuth	1104	60	132	305	+71	−	200	6
290°, dip 21°; mean of	1106	−11	56	313	−56	−	400	3
6 sites; VGP dispersion	1109	20	90	297	13	−	400	1
parameter 5.3, cone of confidence 32.0°; 3 sites rejected	Mean:	27	78	310	12			
HND-10	1110	20	23	354	27	−	400	2
lat 55°8′N, long 199°2′W;	1111	30	105	291	39	+	400	2
thickness sampled 49 m;	1113	25	79	309	11	+	50	1
bedding correction: azimuth	1114	23	51	330	−14	+	100	2
273°, dip 16°; mean of	1117	30	68	319	−10	−	400	28
9 sites; VGP dispersion	1119	39	61	328	22	+	200	21
parameter 10.8, cone of	1121	31	59	327	6	+	100	8
confidence 16.4°; 4 sites	1122	7	89	290	−10	+	200	46
rejected	1124	39	34	348	11	+	400	24
	Mean:	29	64	322	10			

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TABLE 1. (*Continued*)

Area	Site no.	VGP λ	VGP ϕ	D	I	Polarity	AF (Oe)	Intensity (× 10^{-6} emu/cm^3)
SHM-1	1180	8	95	287	− 4	−	100	5
lat 55°2′N, long 200°W;	1181	29	82	308	18	−	100	14
thickness sampled 169 m;	1182	40	60	330	22	−	100	8
bedding correction: azimuth	1183	39	51	336	16	−	100	14
317°, dip 63°; mean of	1184	44	54	335	25	−	100	13
19 sites; VGP dispersion	1185	20	76	309	− 1	−	100	5
parameter 31.4, cone of	1186	37	66	324	20	−	100	8
confidence 6.1°; no sites	1187	33	61	326	10	−	100	21
rejected	1188	34	61	327	12	−	100	22
	1189	35	60	328	12	−	100	13
	1190	42	50	337	20	−	100	17
	1191	33	68	321	14	−	100	20
	1192	33	77	314	21	−	100	28
	1193	32	59	328	7	−	100	20
	1194	35	56	331	11	−	100	15
	1195	40	55	334	20	−	100	46
	1196	24	65	320	− 4	−	100	14
	1197	32	58	329	6	−	100	19
	1198	33	30	352	4	−	100	28
	Mean:	33	63	325	12			
SHM-2	1200	36	91	306	35	−	100	3
lat 55°2′N, long 200°W;	1202	45	51	337	25	−	200	3
thickness sampled 54 m;	1203	44	89	312	44	−	200	3
bedding correction: azimuth	1205	46	88	314	45	−	200	136
350°, dip 28°; mean of	1206	38	87	310	35	−	400	5
8 sites; VGP dispersion	1207	79	138	341	73	−	200	187
parameter 7.5, cone of	1209	57	197	322	88	−	400	93
confidence 21.7°; 3 sites	1210	70	9	5	55	−	200	81
rejected								
	Mean:	59	87	323	51			
IVN	17	54	264	66	70	−	200	1
lat 55°9′N, long 200°5′W;	18	39	254	92	68	−	50	3
thickness sampled 51 m;	20	68	125	321	71	−	200	4
bedding correction: azimuth	21	49	106	305	56	−	200	5
257°, dip 12°; mean of	22	66	189	336	84	−	400	1
8 sites; VGP dispersion	23	40	221	131	80	−	200	4
parameter 4.8, cone of	24	38	110	294	50	−	200	9
confidence 28°; 3 sites	25	45	172	250	79	−	400	3
rejected								
	Mean:	66	183	323	81			
CNB-1	550	26	104	289	33	+	180	1
lat 55°5′N, long 198°8′W;	551	65	115	318	68	+	90	0
thickness sampled ~100 m;	553	43	65	327	29	+	800	2
bedding correction: azimuth	554	46	114	297	58	+	800	1
354°, dip 10°; mean of	555	67	54	344	55	+	800	2
22 sites; VGP dispersion	556	32	67	320	13	−	800	2
parameter 8.9, cone of	557	87	93	355	70	−	800	0
confidence 11°; 2 sites	558	15	139	154	48	−	400	3
rejected	559	55	93	317	55	+	800	1
	560	31	51	333	2	+	400	1
	561	60	213	55	86	−	180	2
	562	39	119	287	56	−	800	15
	563	47	44	342	27	+	800	1
	565	52	73	320	45	+	180	1
	566	41	98	304	46	−	800	3
	567	44	135	281	67	−	400	51
	568	52	99	311	56	+	800	87
	569	34	62	325	14	−	800	69

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strata on Adak Island are in fact Eocene. The original dating was done on the basis of leaf impressions (Coats, 1956); the Eocene age was determined largely from microfossils (Scholl and others, 1969).

The detailed dating for the individual sites is very variable in quality, some units having very good fossil control or radiometric ages and others relying exclusively on lithologic correlations. The basic age controls used in each area are listed in Table 2 and are discussed below. Where an age in millions of years is quoted alongside an epoch or stage name, these are based on the time scale of Van Eysinga (1975).

SAMPLING SITES

Middle Jurassic

Tuxedni Bay: TXD-3, TXD-4, TXD-6, TXD-10. The details of the paleomagnetic measurements for each of these areas are described in Packer and Stone (1974), so only a very brief summary of the main points concerning their ages and mean paleomagnetic directions is given here.

Of these four locations, TXD-3 and TXD-10 were located in the Aleutian Jurassic batholith system, with a potassium-argon age of 168 to 180 m.y. (Reed and Lanphere, 1973). The paleomagnetic data for these two locations have not been included in the final Jurassic mean, even though they comply with the overall selection criteria, because of the difficulties in interpreting data from large intrusive masses (Packer and Stone, 1974).

The other two locations, TXD-4 and TXD-6, are both in sedimentary rocks. Both are well dated on fossil evidence (Detterman and Hartsock, 1966), TXD-6 being Bajocian in age (170 m.y.) and TXD-4 being in the Tonnie Siltstone of the Chinitna Formation, with an age of about 165 m.y. The mean paleomagnetic directions are considered to represent the original magnetic field (Table 2).

Katmai National Monument: GRV-1, GRV-2, KTM-3. These three sampling localities were all in a unit mapped as part of the Aleutian batholith system and radiometrically dated as being 156 to 162 m.y. old (Reed and Lanphere, 1973). The details of the paleomagnetic data are given in Packer and Stone (1974); the overall means and statistics are given in Table 2. Internal inconsistencies in these batholith data make them suspect (Packer and Stone, 1974); thus, they are not included in the overall mean for the Jurassic Period.

It is perhaps of interest to note that the paleomagnetic directions for three of the sampling localities in the batholiths (TXD-10, GRV-2, KTM-3) give mean directions close to the "Asian" Cretaceous mean described below; one (TXD-3) is close to the Jurassic mean, and one (GRV-1) is displaced from either group. This grouping could easily lead to speculation on the time needed for cooling between the temperature at which the radiometric clock is set and the locking in of the magnetic vectors; however, that is beyond the scope of this paper.

Tuxedni Bay: TXD-1, TXD-5, TXD-7, TXD-8, TXD-9 (Upper Jurassic). The paleomagnetic data for all of these localities are described in Packer and Stone (1974). The ages listed in Table 2 are derived from fossil evidence described by Detterman and Hartsock (1966) and show a range from late Callovian (165 m.y.) to latest Kimmeridgian (148 m.y.). All are considered as magnetically stable and representative of the geomagnetic field for Late Jurassic time.

Katmai National Monument: KTM-1, KTM-2, GRV-4. The two KTM localities are from the Naknek Formation near Naknek Lake and are well dated as being Kimmeridgian (approximately 150 m.y.) in age (Burk, 1965).

Locality GRV-4, on the shores of Grosvenor Lake, has been assigned to the Oxfordian stage (approximately 160 m.y. B.P.) by Burk (1965) and others, largely on the basis that it is lithologically similar to the Chisik conglomerate in the Iliamna area and is stratigraphically some 24 m (80 ft) lower than some fossiliferous rocks assigned a Portlandian or Kimmeridgian age. The details of the paleomagnetic measurements are given in Packer and Stone (1974),

TABLE 1. *(Continued)*

Area	Site no.	VGP λ	VGP φ	D	I	Polarity	AF (Oe)	Intensity (× 10^{-6} emu/cm^3)
CNB-1 (continued)								
	570	69	122	322	71	+	400	169
	571	59	96	319	59	−	400	45
	572	58	129	303	69	−	400	43
	573	54	121	302	66	−	800	7.4
	Mean:	54	97	317	55			
CNB-2	574	70	62	342	58	+	190	3
lat 55°5′N, long 198°8′W;	575	19	147	249	56	+	800	57
thickness sampled ~40 m;	576	35	134	273	61	+	800	1
bedding correction: azimuth	577	49	39	346	29	+	400	5
250°, dip 14°; mean of	578	30	88	304	25	+	190	1
10 sites; VGP dispersion	579	61	96	321	60	+	800	6
parameter 5.8, cone of	580	18	113	277	32	+	400	1
confidence 21.9°; 1 site	581	29	165	236	70	+	400	5
rejected	582	63	72	333	55	+	400	6
	583	23	129	267	50	+	800	4
	Mean:	45	113	299	55			

Note: λ = latitude, φ = easterly longitude; D = declination and I = inclination of the equivalent magnetic vector direction; plus = north-seeking polarity, minus = south-seeking polarity. Sites rejected did not meet selection criteria described in text. Cone of confidence of mean of VGP at 95% probability level after Fisher (1953).

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and the mean VGP values and statistics are given in Table 2. KTM-2 has been excluded from the overall means because the mean directions were still moving significantly on progressive demagnetization, and the other two localities do not meet the selection criteria (Packer and Stone, 1974).

Chignik Lake: CHL-1, CHL-2, CHL-3, CHL-4, CHL-5. Burk (1965) mapped the Chignik Lake shoreline area as Upper Jurassic Naknek sedimentary rock; however, the dating is based on a lithologic similarity to the Naknek elsewhere and on the observation that it appears to lie stratigraphically below the nearby Jurassic-Cretaceous Staniukovich Formation. Although this does not give a very close control on the age of these sites, there can be little doubt that the rocks are of latest Jurassic age. The paleomagnetic measurements are described in Packer and Stone (1974); all but one group of sites (CHL-3) satisfy the selection criteria.

Herendeen Bay: HND-1, HND-2, HND-3, HND-8, HND-10 (New Data). Sampling areas HND-1 and HND-2 are both located on the west shore of Herendeen Bay and HND-10 near the tip of Bold Bluff Point. All three localities were mapped by Burk (1965) as being of Late Jurassic age; this conclusion is based on several diagnostic fossils from both Bold Bluff Point and somewhat to the west, near Hoodoo Lake. These fossils include *Buchia concentrica* and belemnite fragments, narrowing the age range for locality HND-10 on Bold Bluff Point to Oxfordian-Kimmeridgian (150 to 160 m.y.). However, the other two localities across the bay are dated exclusively on a lithologic correlation.

The paleomagnetic data for the sites at HND-1, HND-2, and HND-10 are listed in Table 1 and the means in Table 2. As can be

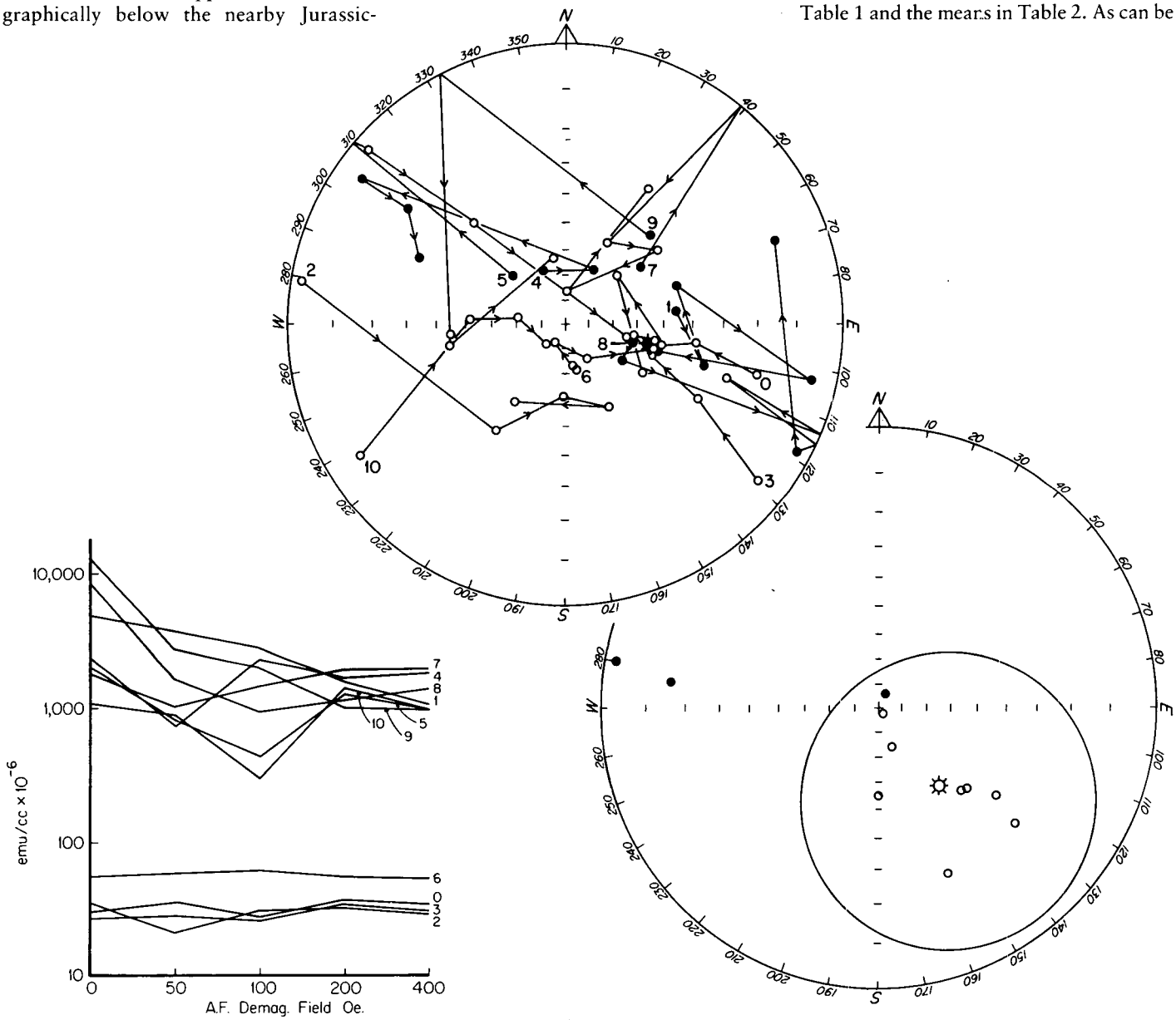


Figure 2. Lower left diagram shows typical demagnetization curves for each site at locality SHM-2. Note logarithimic intensity scale and arithmetic AF level. Top diagram shows uncorrected magnetic vector directions for each site; estimated "closest approach" position as described in text lies near direction labeled 8. Arrows point in directions of higher demagnetization levels. Lower-right diagram shows final vector directions selected using closest approach technique, and their mean corrected to present north and ancient horizontal. All data outside circle of 50° radius are rejected.

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TABLE 2. MIDDLE JURASSIC AREA MEANS

Area	Mean VGP lat λ	Mean VGP long φ	No. of sites	Dispersion parameter k	Cone of confidence (°)	Polarity	AF (Oe)	Age, dating method
Middle Jurassic								
TXD-3	50	336	10	6.5	20.5	+	190	Intrusions, K-Ar, 168 to 170 m.y.
TXD-10	55	79	9	8.3	19.0	+	190	
TXD-6	64	323	25	9.7	9.8	+	190	Red Glacier Formation, Bajocian (170 m.y.)
GRV-1	24	120	10	92.2	5.1	−	190	Intrusions, K-Ar, 156 to 162 m.y.
GRV-2	73	67	13	46.8	6.1	+	95	
KTM-3	72	109	27	3.3	18.4	+	190	
TXD-4*	52	276	9	7.3	20.4	+	95	Chinitna Formation, Tonnie Siltstone (165 m.y.)
Lower Upper Jurassic								
GRV-4	58	96	7	5.6	28.1	+	95	Chisik conglomerate, lithologic correlation, Callovian
TXD-7*	34	295	18	30.2	6.4	+	95	Lower sandstone member, lower Oxfordian
TXD-1	65	0	15	7.3	15.2	+	95	
TXD-9*	18	290	8	21.3	12.3	+	95	Snug Harbor member, Oxfordian-Kimmeridgian
TXD-8*	41	298	13	8.3	15.4	+	95	Pomeroy member, Kimmeridgian
TXD-5*	51	258	22	9.4	10.7	+	95	
Upper Jurrassic–Jurassic-Cretaceous								
HND-10	29	64	9	10.8	16.4	+−	VAR	Naknek Formation, Oxfordian-Kimmeridgian, lithologic correlation with HND-10
HND-1	28	93	12	16.2	11.1	+	100	
HND-2	77	40	7	7.1	24.3	+	VAR	
KTM-1	29	301	24	2.8	21.8	+	95	Naknek Formation, Kimmeridgian
KTM-2	34	115	6	36.7	11.3	+	190	
CHL-1	89	339	10	21.0	10.8	+	95	Uppermost Naknek, lithologic correlation, and stratigraphic relationship to Staniukovich Formation
CHL-2*	49	282	10	7.1	19.5	+	95	
CHL-3	57	297	11	2.8	33.3	+	95	
CHL-4*	15	280	9	8.1	19.3	+	95	
CHL-5*	33	284	12	6.8	18.0	+	95	
HND-3	70	223	12	5.5	20.4	+	VAR	Staniukovich Formation, Jurassic-Cretaceous boundary
HND-8A	25	103	7	10.0	20.1	+	VAR	
HND-8B*	36	254	6	8.5	24.5	+	VAR	
Upper Cretaceous								
PNC-1*	39	9	8	32	9	+	Var	Chignik Formation near basal coal unit (85 m.y.)
CHG-1*	35	318	9	13	13	+	50	Chignik Formation, lithologic correlation with type section
CHG-2*	38	336	8	7	18	+	50	
SHM-1*	33	63	19	31.4	6.1	−	100	
SHM-2*	59	87	8	7.5	21.7	−	Var	
HND-7*	35	91	7	7.4	23.7	+−	400	Chignik Formation, fossil control from HND-4, Campanian
HND-6*	11	92	8	13.5	15.6	−	Var	
HND-4	34	8	8	4.1	31.4	+−	Var	
CNB-1*	54	97	22	8.9	11	+−	Var	Hoodoo Formation, Campanian-Maastrichtian
CNB-2	45	113	10	5.8	21.9	+	Var	
HND-9	27	78	6	5.3	32.0	−	Var	
Lower Tertiary								
PVL-1*	57	286	23	11	9	+−	380	Tolstoi Formation, stratigraphic correlation to type section: macrofossils give Eocene; microfossils give Paleocene
PVL-2*	73	308	13	13	11	+−	285	
CNB-4*	71	336	18	12	9	+−	190	Tolstoi Formation, stratigraphic and lithologic correlation only
IVN	66	183	8	4.8	28	−	Var	Tolstoi Formation, lithologic correlation
VLB-1	76	134	16	10	11	+−	190	Belkofski Formation, stratigraphic below Tolstoi Formation
HND-5	38	288	7	60.2	7.8	+−	Var	Miocene or Cretaceous? (see text)

Note: Localities listed in order of descending age, where possible. Cone of confidence at 95% probability levels of means of λ and φ after Fisher (1953). Plus = north-seeking polarity; minus = south-seeking polarity. AF gives dominant demagnetization level.

* Selected for overall mean.

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seen from both the tables and Figures 3 and 4, the means for these sites of apparent Jurassic age are much closer to the grouping of "Asian" Cretaceous VGP values described below than they are to the Jurassic mean. Both localities show good internal consistency of the data, with no apparent tendency to be smeared out toward the Jurassic magnetic directions, which would argue against a Cretaceous remagnetization. The sites at HND-10 have mixed polarities. The opposite polarities are somewhat divergent from one another but are reversed within the admittedly large 95% confidence limits. Only HND-2 has a mean NRM close to the present field before correction for local dip is applied. In the overall Jurassic mean, sites at HND-1 and HND-10 have been excluded on the basis that they lie almost 90° from the mean of the other sites.

The sites at localities HND-3 and HND-8 were mapped by Burk (1965) as being in the Staniukovich Formation of Jurassic-Cretaceous age. Both localities produced fossils, largely *Buccia (piochii)* and *Inoceramus,* and indicate an age appropriate to the Jurassic-Cretaceous boundary (141 m.y.). The paleomagnetic data from these two areas are listed in Tables 1 and 2. The data from HND-3, although not satisfying the selection criteria for inclusion in the overall Jurassic mean, has a VGP consistent with the other Jurassic data. The data from HND-8 divide into two relatively well-defined groups. The stratigraphically lower of these is consistent with the Jurassic mean, and the other is quite close to the "Asian" Cretaceous VGP. The division in the magnetic directions coincides with a marked change in the intensity of magnetization of the sites (see Table 1), a change in the behavior on demagnetization, and also a change in the type of rocks. No unconformity or other break was noticed in the field, however, nor remarked upon by Burk (1965). Only the mean of the lowermost group has been included in the overall Jurassic mean. The uppermost, possibly Cretaceous group has been included in Table 2 but not included in either mean.

Cretaceous

Shumagin Islands: SHM-1, SHM-2 (New Data). Two locations were sampled in the Shumagin Formation, both on Nagai Island, at Pirate Shake and Sanborn Harbor. Both were sedimentary sections displaying a wide range of rock types, from conglomerates to siltstones. The sites sampled were dominantly in the finer grained material but included some coarse sandstones.

The ages of the two sections are considered to be essentially the same and were given as Jurassic-Cretaceous by Burk (1965), on the basis of *Inoceramus terrebellina* fragments. Moore (1973) assigned a Late Cretaceous age to the rocks on the basis of further work on the *Inoceramus* fossils and also on the lithologic correlation of the Shumagin Formation with the Upper Cretaceous Kodiak Formation. These Late Cretaceous ages were also confirmed by Jones and Clark (1973).

The NRM vectors for the sites at SHM-1 all show a reversed magnetization relative to the present field. On demagnetization most sites showed a slight increase in intensity, presumably due to the removal of a small component along the present field direction, and showed minimum scatter in directions after being exposed to 100-Oe demagnetizing fields.

The direction of the NRM vectors from SHM-2 was in essentially the opposite direction to the Earth's present field; the overall scatter was, however, considerably greater than for SHM-1. On demagnetization, the individual sites reacted differently to the different applied fields, and the "closest approach" technique described earlier was applied. Three of the 11 sites did not approach the mean of the remainder, and they maintained a basically normal magnetization direction. These were excluded from the overall mean, as is shown in Table 2.

Painter Creek and Chignik Lagoon: PNC-1, CHG-1, CHG-2. These three locations and their associated paleomagnetic measurements have been described by Stone and Packer (1977); only the overall mean data are presented here, in Table 2. The Chignik Lagoon samples are from locations very close to the Chignik Formation type section. This section is well dated on fossil evidence as being Campanian (70 to 75 m.y.) in age (Burk, 1965). The Painter Creek section conformably overlies the

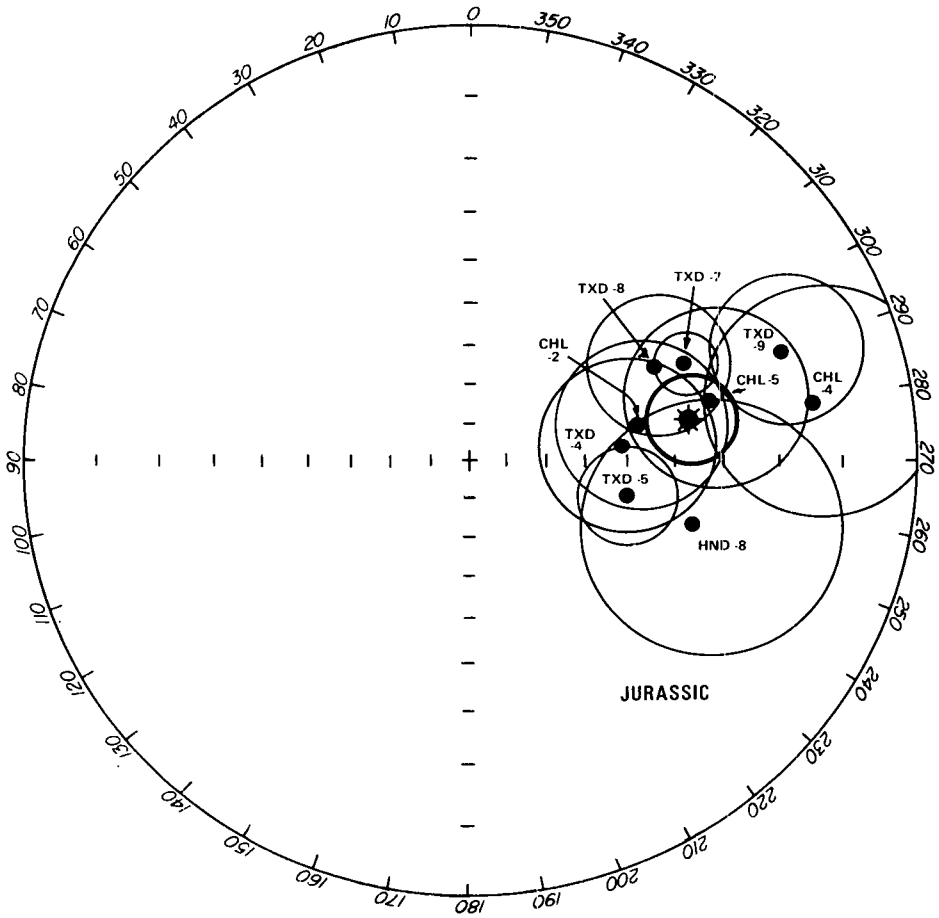


Figure 3. Individual paleomagnetic poles for localities of Jurassic age which satisfy selection criteria, and their mean. Circles of confidence at 95% level. North hemisphere plot.

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basal coal member of the Chignik Formation and is thus essentially the same age as the Chignik Lagoon sections, although it is perhaps a few feet lower stratigraphically.

Herendeen Bay: HND-4, HND-6, HND-7 (New Data). Paleomagnetic collections were made at each of these three areas, which are all in the same general region. The three areas are all in the vicinity of Gull Point in the southeastern arm of Herendeen Bay. All three are beach exposures, areas 6 and 7 being part of a more or less continuous cliff exposure. Area 4 is on the other side of the headland and is considered to be part of the same overall rock sequence on the basis of the similarity of lithology and on general structural trends. Direct evidence of the age of the rocks comes from *Cymatoceras* and *Inoceramus* fossils collected near HND-7, the age of the other two areas being extrapolated from here, and from rocks of similar type exposed across the bay. Burk (1965) assigned a Campanian age to these rocks, and he considered them to be part of the Chignik Formation.

Selection of the data was done on the basis of closest approach of magnetic vectors to an estimated mean, as described earlier. Using this technique, all sites except HND-4 satisfied the criteria and are included in the overall mean; however, several of the sites showed greater than average scatter; therefore, a higher than average number of sites was rejected.

In each of these cases the sites are dominantly reversely magnetized. The demagnetization curves show only a small initial increase in intensity with progressive demagnetizing fields, indicating that the secondary components in the direction of the present magnetic field direction are not large. This demonstrates a certain stability of magnetization. It is of interest to note that even though HND-4 is the only section not included in the final analysis, its mean direction fits with the general groupings of Cretaceous paleomagnetic poles.

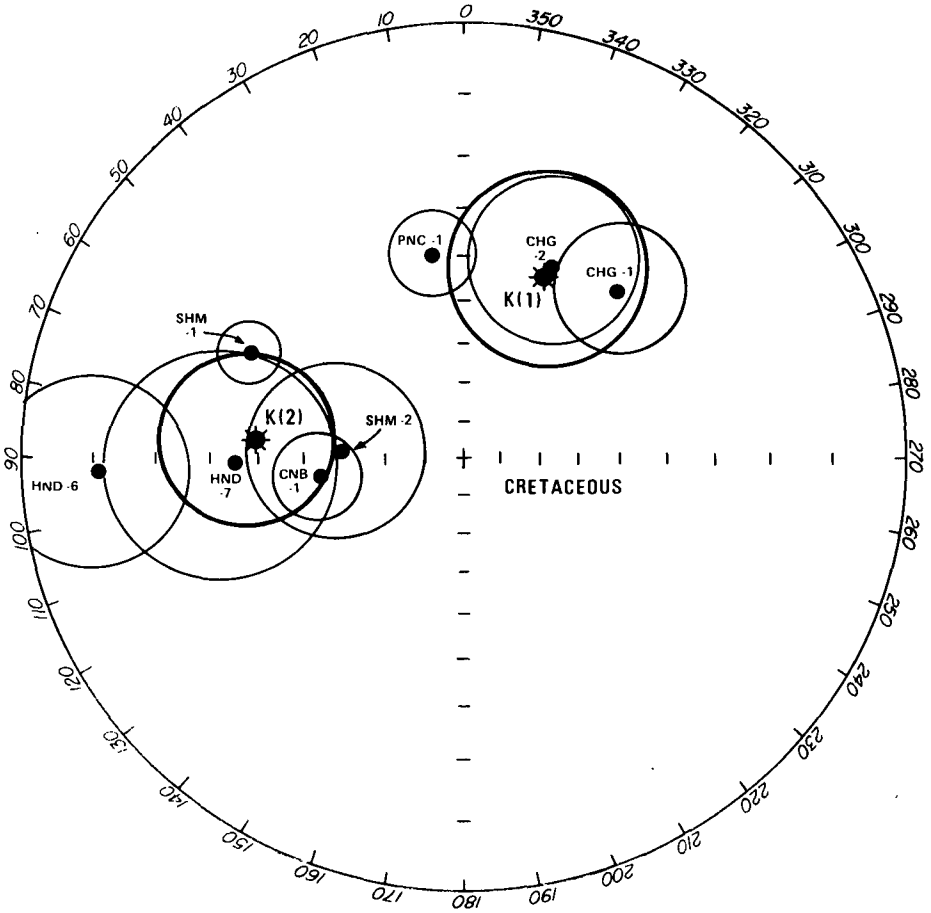


Figure 4. Paleomagnetic poles for localities of Cretaceous age which satisfy selection criteria. These have been subdivided into two groups: "Atlantic" (upper right) and "Asian" (left). Means and circles of confidence are at 95% level. North hemisphere plot.

Herendeen Bay: HND-9 (New Data). This location is near the southeast end of Herendeen Bay and was mapped by Burk (1965) as being in the Hoodoo Formation (approximately 65 m.y. old), presumably on a lithologic correlation, although it is not clear what evidence was actually used. The location is also very close to an area mapped as being of Eocene age, but the boundary between them was not seen in the field. For the purpose of this study, a latest Cretaceous age has been assumed. As can be seen from Table 2, the magnetic directions are indeed consistent with other Cretaceous rocks; however, the overall mean has not been included in the final analysis, because the dispersion of the individual sites was greater than allowed by the selection criteria. All the sites are reversely magnetized, and the NRM directions showed no correlation with the present or axial field directions.

Canoe Bay: CNB-1, CNB-2. These two locations are situated in innermost Canoe Bay off Pavlof Bay. Both appear to be in the same formation but exposed on opposite sides of the bay. The lithology of these very fine-grained and silty sandstones is very similar to that of the Hoodoo Formation at the type locality a few kilometres away. The Hoodoo Formation is considered to be uppermost Cretaceous, and the Canoe Bay sections appear to be stratigraphically beneath the nearby Tolstoi Formation of early Tertiary age. The age of the Hoodoo Formation is based largely on various forms of *Inoceramus*, which place it as Campanian or Maastrichian (65 to 75 m.y.), and the discovery of a large coiled ammonite precludes a significantly post-Cretaceous age (Burk, 1965).

An initial report on the paleomagnetism of the samples from CNB-1 was given in Stone and Packer (1977); however, further demagnetization of these rocks since that time has significantly decreased the dispersion of the data. The rocks show a mixed polarity, the means of each being coincident within the confidence limits. The data from the individual sites and their means are listed in Tables 1 and 2.

The paleomagnetic data from CNB-2 across the bay (Table 1) give directions of magnetization similar to CNB-1, within the confidence limits. These samples were all normally magnetized but represent only 4 m (14 ft) of stratigraphic section. The dispersion of the magnetic vectors of individual sites is marginally too great to allow inclusion in the overall mean.

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Lower Tertiary

Volcano Bay: VLB-1. Burk (1965) mapped the area from which these samples were collected as part of the Belkofski Formation of probable Eocene age. The age of these sites is open to question on two separate arguments. First, the age of the Belkofski Formation is based on a few nondiagnostic wood and leaf fragments and on its stratigraphic relationship with the underlying Tolstoi Formation, whose age is also debatable. Second, the area from which the paleomagnetic samples were taken is mapped as part of the Belkofski Formation only on a lithologic correlation with the type section. The details of the paleomagnetic measurements are described in Stone and Packer (1977); the mean results are listed in Table 2.

Pavlof Bay and Canoe Bay: PVL-1, PVL-2, CNB-4. These three areas were considered by Burk (1965) to be part of the Tolstoi Formation, the type section for which is close by, near the mouth of Pavlof Bay. The Tolstoi Formation is generally considered to be of Eocene age, based on evidence from macrofossils; however, there is some confusion in this, because the plant remains found in the rocks are more consistent with a Paleocene age (Burk, 1965). Details of the paleomagnetic data are described in Stone and Packer (1977); the means and confidence limits are listed in Table 2 and shown in Figure 5.

Herendeen Bay: HND-5 (New Data). The sites in this area, at beach outcrops along the western shore of Herendeen Bay, were mapped by Burk (1965) as of probable Pliocene or Pleistocene age and unlikely to be older than Miocene, on the basis of a pelecypod fauna; however, while collecting samples for the paleomagnetic work described here, an oil company field party was observed to recover fossils of apparent Cretaceous age from the same section (R. C. Allison, 1973, personal commun.). Allison and Addicott (1976) noted that fossils collected slightly inland from the sampling locality give a Miocene age for a section also mapped as Pliocene by Burk (1965). Allison and Addicott also noted that the paleomagnetic sampling locality could be Cretaceous in age but did not specify their reasons for saying so. The paleomagnetic data are listed with other lower Tertiary sites but have not been included in the mean. Selected paleomagnetic data from individual sites from this locality are listed in Table 1.

As shown in Table 1, an unusually high proportion (5 out of 12) of the individual sites was rejected using the criteria outlined earlier; however, the grouping of the selected data shows a very small dispersion, and the rejected samples showed no consistency of magnetic directions relative to each other and in general did not change directions significantly on demagnetization. The low intensities of magnetization precluded further demagnetization. The magnetic polarities of these sites are mixed.

Ivanof Bay: IVN-1 (New Data). The sites at Ivanof Bay are located near the village of Perryville. The rocks samples were mapped by Burk (1965) as being part of the Naknek Formation, but recent work (S. McCoy, 1978, personal commun.) has indicated that they are part of the Tolstoi Formation, of Eocene age. The paleomagnetic data (Table 1) are all reversely magnetized. They give a VGP displaced to the southeast from the means of the other Tolstoi Formation data, but they also have a dispersion that is too large to allow their inclusion in the overall means shown in Table 3.

DISCUSSION OF PALEOMAGNETIC DATA

Jurassic

The VGP values for the various locations have been divided into Middle, lower Upper, and upper Upper Jurassic sequences. As can be seen by comparing the locations of the means (Table 3), the three means are very close together and are not significantly different from the mean of all the Jurassic data combined. This, however, excludes the two divergent Jurassic locations (HND-1, HND-10). It is true that the scatter of directions for the selected Jurassic VGP values is still fairly large (Fig. 3) and, perhaps more puzzling, elongate in a way that is not systematic with age of the site. This elongate distribution could perhaps be explained in terms of a smearing out of apparent pole positions between the present field

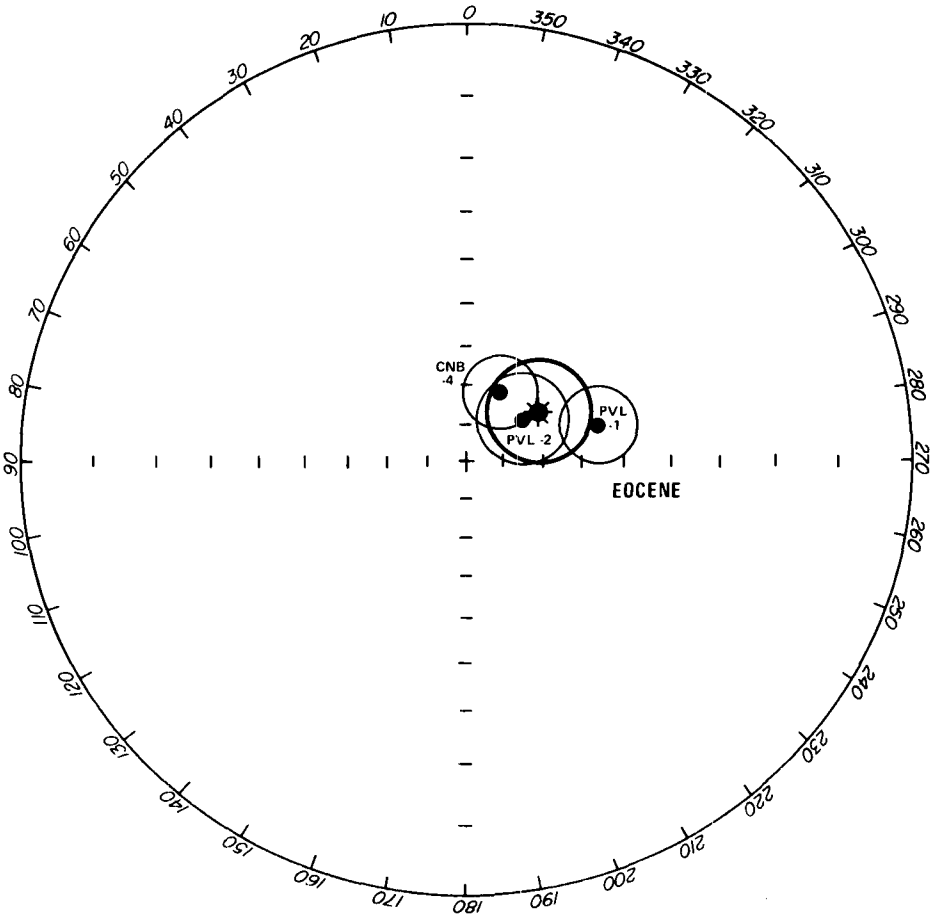


Figure 5. Paleomagnetic poles for localities of Eocene age which satisfy selection criteria, and their mean. Circles of confidence at 95% level. North hemisphere plot.

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and the original direction (incomplete magnetic cleaning). Comparing the scatter of the directions from the sampling localities before and after correcting for local dip, it is found that if all the means that fulfill the selection criteria are used, then the fold test fails. If the area means that are still close to the present field direction after demagnetization are also excluded (TXD-1, TXD-6, HND-2, CHL-1), then the fold test shows no difference between the corrected and uncorrected data. These four area means have been excluded from the final mean on this basis. The distribution of VGP values in this final selection is still elongate, but much less so. The fact that the fold test is only neutral and not positive, combined with the elongate distribution, may indicate that further magnetic cleaning is necessary; however, this will have to await new collections on which thermal demagnetization techniques can be applied. Though the mean VGP obtained as described above is the one used in the speculative reconstruction of Figure 6, it should be kept in mind that the real VGP may indicate an even lower paleolatitude.

Cretaceous

The area means for the Cretaceous, which in practice is entirely limited to the Upper Cretaceous, divide into two groups (Table 3; Fig. 4). One group consists of data from locations on the shores of Chignik Lagoon and from Painter Creek slightly to the north and gives a mean VGP located in the Atlantic Ocean. Another locality (HND-4) in Herendeen Bay gives a pole position close to this group, although the data have a very large dispersion and are not included in the overall statistics. The second group of Cretaceous VGP values contains sites in Herendeen Bay, in Canoe Bay, and on Nagai Island in the Shumagin Islands. The latter group of VGP values, which falls in northern Asia, is the best defined of the two. The means for both groups are included in Table 3 and Figures 4 and 6.

As mentioned above, all the sites are confined to the uppermost Cretaceous and probably do not span more than 15 m.y. or so in toto. This very short time span applies very severe constraints to any proposed mechanism for producing the observed separation of poles by overall plate motions.

In interpreting such data, it would obviously be helpful to have accurate relative ages for the sections sampled. Unfortunately, the fossil-based dating is adequate only for subdividing the Hoodoo and Chignik Formations. No radiometric ages are available. As a result, the only positive age constraint is the fact that the oldest sites, from the Hoodoo Formation, are included in the group of "Asian" poles. The data listed in Table 2 have been assigned relative ages based in part on the paleomagnetic data. The three area means included in the pole located in the Atlantic are all normally magnetized, whereas the other poles are all either reversely magnetized or have mixed polarities. According to the polarity time scale of Watkins (1976) and McElhinny and Burek (1971), essentially everything older than mid-Campanian (74 m.y. B.P.) back to mid-Santonian time (80 m.y. B.P.) is normally magnetized, whereas from about 74 m.y. B.P. into Tertiary time mixed polarities were common. This time scale is consistent with a Campanian age for the "Atlantic poles" and a Maastrichtian age for the others.

TABLE 3. MEAN VGP VALUES FOR SELECTED AREA MEANS FROM SOUTHWEST ALASKA

	Mean of mean VGP		No. of localities	Dispersion parameter, k	Cones of confidence (°)		No. of sites
	lat λ	long ϕ			at 95%	at 63%	
Tertiary	68	305	3	38	13	7	54
Upper Cretaceous							
"Atlantic" group	40	337	3	16	20	12	
"Asian" group	39	85	5	14.0	17	10	
All	52	51	8	3.7	26	15	92
Jurassic							
Uppermost	34	275	4	20	16	9	37
Lower Upper	37	297	4	18	17	10	61
Middle	52	276	1	..	..	..	9
All	37	281	9	21	10	6	107

Note: Cones of confidence after Fisher (1953).

As with the Jurassic data described earlier, a fold test carried out by comparing the scatter of the VGP values relative to present and ancient horizontals showed no significant differences. This was true for the data considered either collectively or as two separate groups.

Though the sites producing the two groups of poles are separated geographically, the separation of the poles is considered real and not a result of local tectonics. The basis for this conclusion is the lack of geologic evidence for a relative rotation of the magnitude required. Countering this, it should be remembered that the geology is not well enough known to be able to positively rule out such a tectonic explanation.

Tertiary

None of the five separate localities of apparent Tertiary age sampled have unambiguous evidence for their assigned ages. The four areas in the Tolstoi Formation (PVL-1, PVL-2, CNB-4, IVN) are probably the least debatable, although, as explained above, the fossils found in the type section do not give self-consistent ages; however, an Eocene age (approximately 40 to 55 m.y.) is the most probable. The other two localities are very debatable, HND-5 because of the unconfirmed report of Cretaceous fossils being found there, and VLB-1 because the dating is based exclusively on a lithologic correlation to another poorly dated section. Only the sites in the Tolstoi Formation that satisfy the selection criteria have been included in the mean.

As can be seen from Table 3, the cones of confidence about the final period means are relatively large. This is in part due to a decision to use the small number of means of areas as opposed to the more numerous site means. As explained earlier, the terminology of McElhinny (1973) has been followed, inasmuch as a site is taken to represent a unit of time. In the sampling pattern used for this study, samples were taken from individual beds in the sedimentary sequences and are therefore considered as representing a time unit; however, by sampling a succession of beds at one location, then another succession of beds of the same general age at another location, the distribution of samples through time is very unlikely to be uniform. Because of this, it was decided that the mean of area means was more likely to give a meaningful representation of the original field than combining individual site means. One of the results

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of employing this technique is the small number of vectors used for the final means, and the consequent large circles of confidence.

INTERPRETATION

The paleomagnetic data described above can be interpreted in many different ways, depending on the assumptions made. We have made the common assumption that in all cases we have been measuring the magnetic field direction of an ancient geocentric dipole and that the time represented by the mean for each area is long enough to average out any nondipole components. On this assumption, there can be only one paleomagnetic pole for any given time. If the poles for two different areas are significantly displaced from one another, then this means that the areas have undergone relative movement. The relative paleolatitudes can be obtained by simply superimposing the poles. Some further constraint is needed to determine the relative paleolongitude. To interpret the paleomagnetic data described above, we have generally constrained southwestern Alaska to be part of North America in a geologic sense, so that they must always be essentially in contact. This constraint does, however, lead to conflicts with the geology in Jurassic time.

For the speculative reconstruction of the paleogeography of southern Alaska relative to North America, an apparent polar wander path has been drawn, as shown in Figure 6, so that one end is at the mean Jurassic pole and the curve then cuts the circles of confidence of both the Cretaceous poles and also passes through the mean of all the Cretaceous data. It then loops back to cut the circle of confidence of the Tertiary pole and on to the present geographic axis.

The polar wander curve for North America is drawn from the data listed in Table 4. The various positions of the block representing the sampling area in southwestern Alaska have been derived exclusively by matching the appropriate parts of the two apparent polar wander paths shown.

As can be seen from Figure 6, a southward movement of the block representing southwestern Alaska is required between Jurassic and Cretaceous time. This sense of motion does not fit easily into current models of Pacific plate motions, but it is not excluded by them. The lack of detailed information about the relative motions along the east Pacific margin at that time precludes resolution of the problem on these grounds. An argument against the relative position of southwestern Alaska and North America shown for Jurassic time is that the Alaskan Jurassic batholiths, shelf sediments, deep-water sediments, and possibly blueschist localities all point to a Jurassic subduction zone on what is now the southeast side of the Alaska Peninsula. As can be seen from the reconstruction plotted in Figure 6, this would require an active spreading center and trench system between the two blocks. The paleomagnetic data do not allow a rotation of the block through 180° to get the fossil subduction zone on the "outside" of the system, unless one reconstructs using the south VGP, in which case Jurassic Alaska would be in the Southern Hemisphere. There is a possibility, however, that the evidence for subduction on the southeast side is being misinterpreted and that the present relative positions of the geologic zones are due to en echelon movement of various slivers of material. This argument derives from the fact that each of the terrains mentioned above is fault bounded, and the blueschist localities are considerably older than the batholiths (Forbes and Lanphere, 1973). Unfortunately, none of the explanations is satisfying.

Another factor in this argument is that if the Jurassic subduction zone was indeed on what is now the northwest side of the Aleutian batholiths, and these were part of the subduction zone–batholith line of the western edge of North America, then there should be a gap in the western American

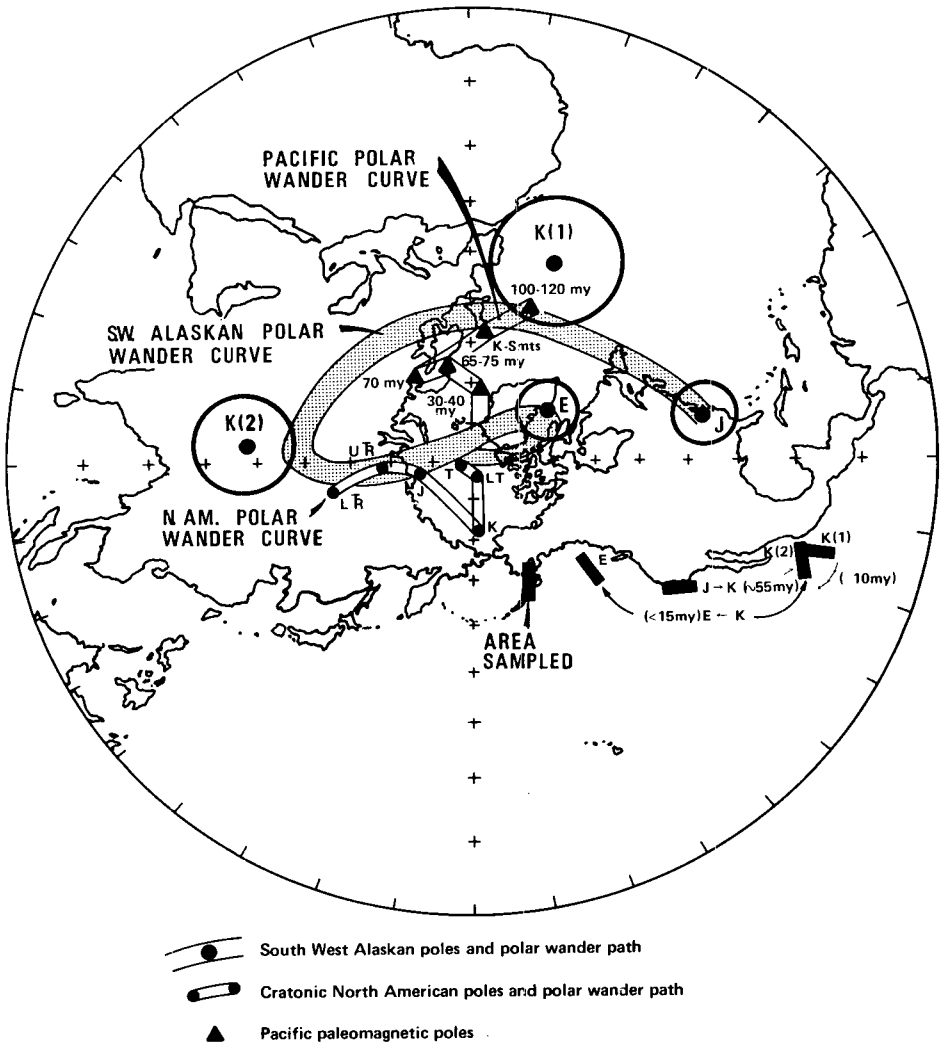


Figure 6. Paleomagnetic poles and polar wander curves for North America, southwestern Alaska, and Pacific. Note that circles of confidence for Alaskan poles are at 63% level for clarity. Reconstructed position of Alaskan sampling area was obtained by superimposing Alaskan and North American poles and constraining sampling area to be in contact with North America.

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system into which the Alaska Peninsula batholiths could fit. One could postulate that the apparent gap north of the Klamath Mountains originated in this way. Similarily, the paleomagnetic data at its error limits (at the 95% confidence level) allow Jurassic southwest Alaska to be near the disturbed zone at the southern end of the west coast batholith line.

The long section of polar wander curve between the two Alaskan Cretaceous paleomagnetic poles, as sketched in Figure 6, requires a simple rotation of the Alaskan block with no translation relative to North America. If this is real, then it places some geometric limits on the size of the block that can have been involved and also resolves the problem of how to create such a long polar path in a very short time. It is of interest to note in this context that the axis of rotation required by the Alaskan paleomagnetic data is within 20° of the Euler rotation pole for the Pacific relative to the hot spots as defined by the Emperor Seamount chain (Clague and Jarrard, 1973). The location of the Euler pole relative to North America cannot be located exactly because it depends on the assumptions made concerning the relative motion of the Pacific and North American plates.

A polar wander curve drawn through the paleomagnetic poles for the Pacific plate (Table 5), as derived from both the seamount data (Harrison and others, 1975) and from marine magnetic anomalies (Vine, 1968; Larson and Chase, 1972; Cande, 1976), is remarkably similar to the Alaskan apparent polar wander curve, as can be seen from Figure 6. The implication of the apparent rotation of Alaska and the similarity of the Pacific and Alaska polar wander curves, if taken at face value, is that in Cretaceous time the southwest Alaskan block was part of the Pacific, attached to either the Kula or Farallon plates. If this is true, then a rapid northward motion of the Alaskan block would be expected in line with the motion proposed for the Pacific plate (see, for instance, Clague and Jarrard, 1973). That such a motion occurred is also implied by the Alaskan Tertiary paleomagnetic data, which place the Alaskan block much farther north in Eocene time. This is a larger amount of translation in a somewhat shorter time than is generally proposed for the Pacific plate. It is true that the dating on the Tertiary material is not good, but even making it as old as possible allows only about 25 m.y. for about 40° of translation, or roughly 16 cm/yr. This is not an impossible figure, and the explanation of the high velocity may lie in the fact that the Alaskan block was probably on the Kula plate (Atwater, 1970) at this time, so it is combining the Kula Ridge spreading rates with those representing Pacific plate motion.

TABLE 4. NORTH AMERICAN PALEOMAGNETIC POLES

	lat λ	long ϕ	No. of mean poles	Dispersion parameter, k	Cone of confidence (°)	Reference
Upper Tertiary	87	140	9	56	7	McElhinny (1973)
Lower Tertiary	85	197	6	23	14	McElhinny (1973)
Cretaceous	69	187	10	223	3.2	Stone and Packer (1977)
Jurassic	77	107	7	23	11	Packer and Stone (1974)
Upper Triassic	68	97	11	61	6	McElhinny (1973)
Lower Triassic	56	104	8	89	6	McElhinny (1973)
Permian	46	117	6	86	7	McElhinny (1973)

Note: Cone of confidence at 95% level after Fisher (1953).

TABLE 5. PACIFIC PALEOMAGNETIC POLES

	lat λ	long ϕ	No. of mean poles	Dispersion parameter, k	Cone of confidence (q)	Reference
California seamounts (33–40 m.y.)	71	354			13	Francheteau and others (1970)
Alaskan magnetic bight (65–75 m.y.)	65	13			~10	Vine (1968)
Magnetic lineations (70 m.y.)	66	33			..	Cande (1976)
Cretaceous seamounts (Late Cretaceous)	58	350	26	35	4.8	Harrison and others (1975)
West Pacific lineations (100–120 m.y.)	50	330			..	Larson and Chase (1972)

Note: Cone of confidence at 95% level after Fisher (1953).

CONCLUSIONS

The paleomagnetic data alone, when interpreted using the dipole field hypothesis and constraining southwestern Alaska to be adjacent to North America at all times, places southwestern Alaska at approximately the California-Oregon border in Jurassic time. Sometime between latest Jurassic time and latest Cretaceous time the Alaskan block moved south. During latest Cretaceous time it remained in essentially one location, but rotated through about 90°. From the end of Cretaceous to approximately Eocene time, the Alaskan block sped northward and rotated back to a position approximately parallel to the continental margin off British Columbia. Between Eocene time and the present the block underwent a clockwise rotation to form the present Gulf of Alaska. It must be emphasized that this scenario is based exclusively on the paleomagnetic data and that the error limits allow considerable variations in the timing and positions of the Alaskan block, although the basic sense of motion must remain the same.

In terms of the overall geology, the paleomagnetically based model would predict that the Alaskan block was part of the western North American subduction zone in Late Jurassic time, but with subduction occurring from what is now the northwest side. In Cretaceous time it was fixed to the North American block, but in a more southern latitude, with the relative motion between it and the ancestral Pacific plate being such that the Cretaceous flysch deposits (Moore, 1973) were being formed in the ever-widening gap left by the dextral rotation of the block. Toward the end of Cretaceous time, the northward movement as part of the Kula or Farallon plates began, presumably along a transform boundary on the western edge of the North American plate. With the subduction of the triple junction associated with the Kula Ridge, the relative motion of the northern Pacific changed, and the Alaskan block swung around to its present position.

Because this story is built exclusively on paleomagnetic data from the Alaska Peninsula–Shumagin Island areas, only the

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relative motion of this block has been considered. The size of the block involved in these proposed motions is open to debate, but it seems likely that the block itself was relatively small. It also seems probable that if such motions occurred, then there may have been many such blocks involved. An analysis of the paleomagnetic data from the western edge of North America shows a systematically increased dispersion, when compared with equivalent data from stable or cratonic North America, and also shows a tendency toward more southerly paleolatitudes (Beck, 1976; Stone, 1977). A geologic model for the western edge of North America which can account for much of the geology observed today involves many slivers of continental material being rotated and moved northward along the plate boundary.

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